

Surya Prakash Murugavvel

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Education

Georgia State University

Atlanta, GA

Bachelor of Science in Computer Science, Minor in Mathematics

Expected Graduation: May 2027

- Dean's List: Spring 2025, Summer 2025
- **Relevant Coursework:** Data Structures & Algorithms, System-level Programming, Computer Organization & Programming, Software Development, Machine Learning, Programming Language Concepts, Computer Networks, Discrete Mathematics, Calculus I/II/III, Probability & Statistics, Linear Algebra, Numerical Analysis I, Optimization, Applied Combinatorics, Physics I/II, Electronics

Technical Skills

Languages: C, C++, Python, JavaScript, HTML/CSS, VHDL/Verilog, MATLAB, MySQL, MongoDB

Frameworks/Libraries: ReactJS, Next.js, Three.js, TensorFlow, PyTorch, Pandas, NumPy, Selenium

Networking & Systems: Wireshark, TCP/IP, UDP, DNS

Tools/Technologies: Amazon Web Services (AWS), Docker, Git, GitHub, Emscripten, REST APIs, Blender, Circuit Analysis (DC Circuits, AC Circuits, Fourier Analysis), Oscillators

Platforms: Windows, macOS, Linux

Projects

SecretSociety | *Python, TCP Sockets, Threading, JSON*

November 2025

- Built a **multi-threaded** IRC-inspired chat server and client in Python, supporting concurrent connections via **TCP socket programming** and a structured **JSON-based protocol**
- Implemented **thread synchronization** to manage shared channel state and prevent **race conditions** across concurrent client sessions
- Designed server-side command processing, activity logging, and graceful shutdown to simulate production-grade network service behavior

Style.Me | *Python, JavaScript, Selenium, OpenAI API, Pinterest API*

November 2025

- Built and deployed a Chrome extension that performs real-time style analysis for personalized apparel recommendations from online retailers
- Optimized data collection by integrating **Selenium-based scraping** with fashion APIs to improve **product retrieval speed and coverage**
- Leveraged **LLM-based natural-language processing** to interpret style queries and refined skills in **API integration, data parsing**, and **user-experience** optimization

Threadoku | *C++, JavaScript, WebAssembly, Emscripten, CPR, nlohmann/json*

April 2025

- Architected a multi-threaded Sudoku solver compiled to WebAssembly for **interactive browser** visualization
- Increased algorithmic efficiency by **63%** through parallel processing, thread synchronization, and dynamic load balancing
- Enhanced expertise in **concurrency, performance tuning, and scalable system design** for compute-intensive applications

PaperKeys | *ReactJS, JavaScript, MediaPipe*

February 2025

- Collaborated on an interactive virtual piano that uses **hand tracking** with a printed keyboard for **real-time note feedback**
- Improved tracking precision by replacing OpenCV with custom anchor-point alignment, reducing note-detection errors by **>20%**
- Refined expertise in **computer vision, real-time data processing, and user-interaction modeling** within team-based agile development

DiagnAI | *Python, HTML/CSS, JavaScript, Anthropic API*

October 2024

- Collaborated in an agile team to create an AI-powered health assistant providing personalized symptom screening and medical insights
- Optimized backend asynchronous request handling and knowledge-base integration to reduce query latency by **35%**
- Learned to architect **scalable backends**, manage **asynchronous requests**, and design **intuitive conversational UIs**

Extracurriculars

Rocket Technologies GSU

August 2024 – Present

Software

Atlanta, GA

- Contributed to the design and simulation of model rockets and autonomous rovers using thrust vector control systems to enhance flight stability and precision
- Improved thrust-vectoring performance by **25%** through iterative testing, control-loop tuning, and data analysis with **MATLAB** and embedded software tools
- Gained hands-on experience in **aerospace simulation, sensor calibration, embedded control systems**, and **team leadership** through multidisciplinary collaboration